

PROTEOLYSIS OF HISTIDINE-RICH GLYCOPROTEIN IN PLASMA
AND IN PATIENTS UNDERGOING THROMBOLYTIC THERAPY

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ABSTRACT

Plasmin and kallikrein but not thrombin cleave purified histidine-rich glycoprotein (HRG), and heparin binding inhibits the proteolysis of HRG. To assess the proteolysis of HRG in plasma, immunoaffinity chromatography was used to isolate HRG from human plasma samples and the extent of protein cleavage was determined after electrophoresis under reduced, denaturing conditions. In blood drawn into streptokinase or into urokinase, HRG (78 kDa) was degraded producing peptides ranging in apparent molecular weight from 67 to 9 kDa. In patients undergoing thrombolytic therapy almost no intact HRG remains after 30 minutes, but the levels of circulating HRG are unchanged, indicating that cleaved HRG is not quickly or extensively removed from the circulation.

INTRODUCTION

Thrombolytic therapy of conditions such as venous thrombosis, occlusive vascular disease and disseminated intravascular coagulation utilizes plasminogen activators like tissue plasminogen activator (TPA), streptokinase (SK) and urokinase (UK). Although these agents have long been in clinical use, the side effects of this therapy have yet to be fully assessed.

Recent evidence suggests that histidine-rich glycoprotein (HRG) is an anti-fibrinolytic agent which acts by binding to the primary lysine-binding site of plasminogen (1). Interestingly, HRG binds to heparin counteracting its anticoagulant properties (2,3) and alleviates the inhibition by heparin of thrombin-induced platelet aggregation in vitro

Key words: Histidine-rich glycoprotein, thrombolytic therapy, plasmin, immunoaffinity chromatography

(4). Since HRG is cleaved by plasmin in vitro (5,6), increased plasmin activity caused by thrombolytic therapy could affect circulating HRG with possible deleterious consequences.

Immunoelectrophoresis (5) and immunodiffusion (see below) with polyclonal anti-sera and gel permeation chromatography (7) do not distinguish proteolytically-cleaved HRG from the intact protein in plasma. Therefore, in this work immunoaffinity chromatography was used to isolate HRG from plasma and electrophoresis under reduced, denaturing conditions to demonstrate the proteolysis. The results presented here are the first to determine the extent of proteolytic cleavage of HRG in normal plasma in vitro and in vivo in patients undergoing thrombolytic therapy.

MATERIALS AND METHODS

Plasma was obtained from healthy volunteers after informed consent and approval of the Institutional Review Board for human investigation. Blood was routinely drawn into a cocktail of protease inhibitors containing kallikrein inhibitor (Antagosan, Behring, 400 units/ml final concentration), benzamidine (10 mM), EGTA (10 mM) and 10 mM sodium citrate. Red cells were removed by centrifugation at 2000 g for 20 minutes at 4°C. In experiments to assess the effects on HRG of increased plasmin activity or of heparin, blood samples were drawn into SK (500 units/ml plasma) or into UK (20 units/ml plasma) with and without heparin (4.2 units/ml plasma) and incubated on ice for 1 hour. The plasma samples were frozen in aliquots in liquid N₂ and stored at -70°C until analyzed.

The three patients investigated received streptokinase therapy as an emergency measure to remove clots obstructing major coronary blood vessels. The clots were visualised before infusion and the patency of the previously occluded vessels established after infusion by routine angiograms. All three patients had a previous history of myocardial infarcts and therapy was begun within 8 hours of the onset of chest pain. After informed consent was obtained, the patients received heparin (5000 units) to maintain patency of the catheters and a blood sample was taken before a bolus injection of SK (pre-SK sample) followed by an infusion of SK at a flow of about 5,000 units per minute. Immediately after the infusion of SK the second blood sample was taken (post-SK sample). Patient 1 was a 70 year old male who received 20,000 units bolus followed by 108,000 units infused over 45 minutes. Patient 2 was a 53 year old male who received 90,000 units bolus, 300,000 units infused into the right coronary artery followed by a further 100,000 units bolus and 400,000 units infusion at a second clot site in the left coronary artery. Patient 3 was a 66 year old female who received 20,000 units bolus and 300,000 units infused over 60 minutes. Blood samples were drawn into the cocktail of inhibitors described above and the plasma frozen in aliquots in liquid N₂ and stored at -70°C until analyzed.

Human HRG was isolated by chromatography on phosphocellulose (8) or on carboxymethylcellulose (9). The purity of HRG preparations was assessed by slab gel electrophoresis in the presence of sodium dodecylsulphate (SDS) (10) using 3-17% acrylamide gradients or 10% acrylamide. The reducing agent dithiothreitol (DTT) was included to monitor the integrity of the protein which is extensively disulphide-linked. The concentration

of purified HRG was assessed using an $E_{1\%}^{1\text{cm}}$ of 5.85 at 278 nm (11). Iodination of purified HRG with ^{125}I was carried out using Iodobeads (Pierce), under conditions that produced incorporation of one I per HRG. Labeling of HRG, human serum albumin and human plasminogen with ^{14}C - H_2CO was accomplished by the NaCNBH_3 method (12).

Mancini (13) and Ouchterlony (14) immunodiffusion techniques and preparation of the IgG-BioGel CM-A immunoaffinity chromatography medium (15) were carried out using monospecific polyclonal goat antibody to human HRG. HRG levels in plasma were measured by the Mancini method as described (22). Protein eluted from the immunoaffinity column by thiocyanate was extensively dialyzed against water and lyophilized before electrophoresis. The column was routinely rinsed with buffer and thiocyanate between samples to remove any residual protein. Human HRG isolated by immunoaffinity chromatography was subjected to polyacrylamide electrophoresis (10) under reduced denaturing conditions using 10% or 3-17% gradient polyacrylamide slab gels. Protein bands were detected after electrophoresis by staining with Coomassie blue R250 or silver (16). The "Western" blot procedure (17) was used to demonstrate that all the protein bands in the thiocyanate-eluted fractions, after transfer to a nitrocellulose filter, contained antigenic determinants recognized by the antiserum produced in this laboratory. The only exceptions are two faint bands observed in some samples that were identified as heavy (55 kDa) and light (30 kDa) chains of goat IgG. The blotting procedure was not used with unfractionated plasma since the large amount of albumin present relative to HRG obscures certain fragments of HRG.

Plasminogen was isolated from human plasma (18) and activated either with UK (Calbiochem) or SK (KABI) as described (19). Amidolytic activity was measured using the chromogenic peptide substrate, S2251 (KABI). Thrombin (Sigma) activity was generally 3U/ug protein, and kallikrein (Calbiochem) activity was 0.5U/ug protein. Limited proteolytic digestions of HRG were carried out at 37°C in 15 mM phosphate buffer, pH 7.4 containing 150 mM sodium chloride using 1:10, 1:30, 1:50 and 1:100 ratios by weight of protease to HRG. Incubating purified HRG with SK or with plasminogen alone had no effect. Heparin was added where indicated 20 minutes before the enzyme. The reaction was stopped by addition of electrophoresis solubilisation buffer containing 0.5% SDS and 10 mM DTT and heating at 100°C for 5 minutes.

RESULTS

Purified HRG is susceptible to proteolysis by plasmin in vitro (5,6), but little is known of the sensitivity of HRG to other proteases and of its cleavage in plasma. As a first step the sensitivity of purified HRG to plasmin, thrombin and kallikrein was investigated since these three enzymes are all serine proteases involved in hemostasis. Time course studies of the action of plasmin on purified HRG (Figure 1) reveal that a 9 kDa non-disulphide linked peptide is first cleaved from the 78 kDa native protein producing a 67 kDa species followed by the slower appearance of the 60, 48, 40 and 30 kDa peptides which are the histidine-proline-glycine-rich and -poor peptides previously reported (6) (Figure 5, tracks 1 to 4). Thrombin does not act on HRG (Figure 1, tracks 5 and 6). In contrast, kallikrein extensively cleaves human HRG (Figure 1, tracks 7 and 8) producing a 48 kDa peptide that is somewhat protease-resistant but finally degraded by kallikrein (Figure 1, track 8).

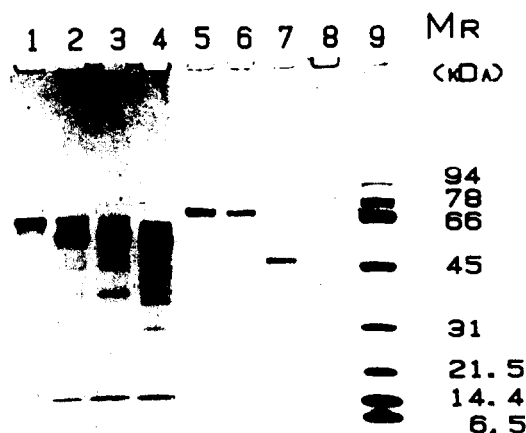


FIG. 1

Effects of plasmin, kallikrein and thrombin on human HRG.

Purified HRG was treated with plasmin (1:50 w/w) at 37°C for 0 (track 1), 5 (track 2), 15 (track 3), and 60 minutes (track 4). HRG was also incubated with thrombin (1:50 w/w) for 15 (track 5) and 60 minutes (track 6), and treated with kallikrein (1:100 w/w) for 2.5 (track 7) and 15 minutes (track 8). Track 9 contains molecular weight standards.

Heparin is bound tightly by HRG (2,3,4) and might thus be expected to affect the proteolysis of HRG. To examine this, heparin-HRG complexes were formed by incubating HRG with 1.5 or 10 molar equivalents of heparin before addition of plasmin (Figure 2, tracks 2 and 3, respectively). Proteolysis of HRG was inhibited in the presence of heparin although plasmin activity, measured with S2251, was unaffected. This result suggests that formation of the tight heparin-HRG complex protects some plasmin-sensitive sites on HRG. Heparin has been reported to inhibit the amidolytic activity of plasmin (20,21), but only at concentrations of S2251 above the K_m .

Since HRG is exposed to these enzymes in the complex milieu of plasma, it is important to assess cleavage of HRG in the more natural setting; and an immunoaffinity chromatographic technique used in this laboratory to examine metal binding in plasma (15) was adapted to this purpose. As shown in Figure 3, over 80% of 50 ug of purified HRG applied to the anti-HRG immunoabsorbent column is retained specifically and is eluted by NaSCN. Neither human serum albumin nor human plasminogen is bound by the column. Fifty ug corresponds to the HRG present in about 0.5 ml of plasma (24). Although the immunoaffinity columns are remarkably stable and can be used up to fifty times, occasionally faint protein bands of 55 kDa and 30 kDa were observed on gels. Control experiments established these bands to be the light and heavy chains of goat IgG. No significant differences in the recoveries ($80 \pm 10\%$) of intact or plasmin-cleaved ^{125}I -HRG or ^{14}C -HRG from the immunoaffinity column were found. Autoradiography after electrophoresis and "Western" blots (17) of both forms of the protein recovered from the column confirmed that the insolubilized antibody recognized both equally.

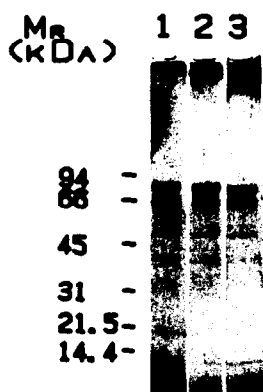


FIG. 2 - left

Effect of heparin on proteolysis of human HRG by plasmin.

Purified HRG was incubated in the absence (track 1) or presence of heparin (1.5 and 10 equivalents, tracks 2 and 3, respectively) for 20 minutes before addition of plasmin (1:50 w/w plasmin:HRG). After 60 minutes proteolysis was stopped by the addition of solubilisation buffer and heating.

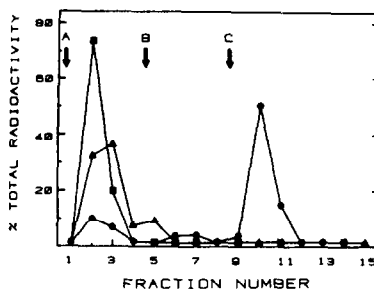


FIG. 3 - right

Immunoaffinity chromatography of human HRG.

Samples were applied to a column containing 3 ml of goat anti-human HRG IgG coupled to CM-BioGel A at arrow A. The column was washed with 5 ml 25 mM Hepes, 0.15 M NaCl, pH 7.4, and with 2 ml Hepes buffer, pH 7.4, containing 1.4 M NaCl (arrow B). Specifically-bound protein is eluted with 2 ml 3.5 M NaSCN (arrow C). Shown are: 50 ug ^{14}C -HRG (circles), 50 ug ^{14}C -human serum albumin (squares) and 50 ug ^{14}C -human plasminogen (triangles). Each protein was run individually and fractions collected and radioactivity determined. Data are presented as % total radioactivity recovered in each fraction. The flow rate was 30 ml/h.

The effects of plasmin activation and heparin binding on HRG was examined in blood drawn from healthy subjects directly into SK or UK, in the presence or absence of heparin. The HRG in these samples was compared with HRG from parallel samples drawn into a cocktail of protease inhibitors. Increasing plasmin activity by the action of SK or UK caused a rapid degradation of HRG (Figure 4A), and the pattern of cleavage products resembles that produced by plasmin (Figure 1). However, using the Ouchterlony technique no immunological differences could be detected between HRG in serum or plasma, between cleaved or intact purified HRG, and between immunoaffinity or regularly purified HRG (Figure 4B). Moreover, no change in HRG concentration with any treatment was noted with the Mancini technique (not shown). These data demonstrate that standard immunodiffusion methods do not distinguish intact from proteolytically cleaved HRG. When blood is drawn into tubes containing both heparin and SK, HRG was protected to some extent from plasmin cleavage even in plasma where several heparin binding proteins exist (Figure 4A, track 3). HRG does not activate plasminogen and is completely stable in its presence (Figure 4A, track 4).



FIG. 4

Characterization of human HRG isolated by immunoaffinity chromatography .

Panel A shows the electrophoretic pattern of human HRG immunoaffinity purified from: plasma alone (track 1); plasma containing SK (track 2); and plasma containing heparin and SK (track 3). Track 4 shows purified HRG mixed with plasminogen, and track 5 plasminogen alone. Incubations were at 4°C for 1 hour. Polyacrylamide gels (10%), containing SDS and DTT were used. The numbers represent the molecular weights of the standard proteins used: phosphorylase a, transferrin, bovine serum albumin, ovalbumin, carbonic anhydrase, soy bean trypsin inhibitor, lysozyme, and Antagasan.

Panel B shows a double-diffusion analysis of the reaction of anti-human HRG (center well) with human plasma from blood drawn into: heparin alone (well 1); into heparin with UK (wells 4), or with SK (well 6); into Antagasan (well 2); into UK (well 3) or SK (well 5). Wells 2, 5 and 6 are aliquots of samples shown in tracks 1, 2 and 3, respectively, of Panel A.

HRG is cleaved rapidly *in vivo* during SK therapy as shown as the almost complete loss of 78 kDa intact HRG in samples taken from patients after SK-infusion (Figure 5). The pattern of peptides (Figure 5, track 2) again resembles the pattern generated by cleavage of HRG by plasmin (Figure 1, track 2), providing the first evidence for an interaction of plasmin with HRG *in vivo*. Nonetheless, the proteolysis of HRG during thrombolytic therapy produced no detectable change in HRG level over 30 to 60 minutes in any of the patients as measured by the Mancini technique, indicating that clipping does not cause immediate or extensive removal of HRG from the circulation. Interestingly, the levels of HRG in all three patients (0.17 ± 0.016 mg/ml, mean $\pm$ standard deviation) were higher than normal controls (0.125 ± 0.03 mg/ml, ref.22), consistent with the elevated levels of HRG in several patients with cardiovascular disorders (22) and myocardial infarction with deep vein thrombosis (23). Whether HRG levels and HRG breakdown can provide additional means of assessing thrombolytic disorders and thrombolytic therapy remains to be determined. Since heparin slows cleavage of HRG *in vitro* (see above), the proteolysis seen here might well be exceeded in patients not receiving concurrent heparin. The two minor bands at 55 and 30 kDa in Figure 5 are from IgG leaked from the column.

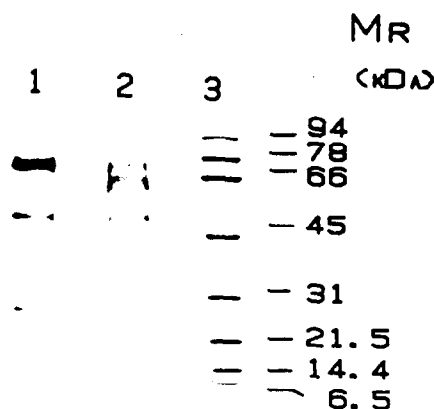


FIG. 5

Effects of thrombolytic therapy on HRG
HRG was purified by immunoaffinity chromatography from samples of plasma drawn before (track 1) and 60 minutes after (track 2) SK infusion and electrophoresed on 3-17% polyacrylamide gels containing SDS and DTT. Track 3 contains molecular weight standard proteins. Essentially the same pattern was seen with all three patients.

DISCUSSION

Serum HRG levels are altered in certain physiological and pathological states (22-26), but the mechanism of alteration and biomedical consequences remain unknown. As a first step in defining the interaction of HRG with proteases and the physiological consequences of proteolytic degradation of HRG, the studies presented here were carried out. The results show that heparin diminishes proteolysis of isolated HRG and of HRG in plasma and that another serum protease, kallikrein, degrades HRG. Proteolysis of HRG was obtained in plasma drawn into SK or UK using a immunoaffinity technique. Importantly, SK therapy was shown here to produce extensive proteolysis of HRG *in vivo*. Plasmin-cleaved HRG is more rapidly cleared from the circulation than the native protein in rabbits (27), suggesting that increased catabolism of HRG might occur in patients undergoing thrombolytic therapy. However, the reported shorter plasma survival time of HRG in patients treated with SK (7) is difficult to relate to the work presented here since intact HRG was not used in the previous study. We observed that almost no intact HRG remains after 30 minutes of SK infusion, but total HRG levels are not significantly altered indicating that cleaved HRG is not immediately or extensively cleared from the circulation. Perhaps several cleavage sites must be hit to produce an effect. Prior to the present study no method to distinguish intact from cleaved HRG in plasma has been reported. Since clipped and intact HRG react identically with polyclonal anti-sera, immunological methods routinely used to examine protein plasma levels do not distinguish the forms.

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